

CHANG HE (Aaron)

Undergraduate, UNSW Sydney · Sydney, Australia

aaron.he@student.unsw.edu.au · aaronhnoraa.github.io · github.com/AaronHnoraA · wiki.pwo101.top

EDUCATION

University of New South Wales (UNSW)

Sydney, Australia

Bachelor of Advanced Computer Science (Honours) (UNSW 3779)

Term 3, 2024 – Present

- Expected graduation: Term 3, 2028. Transferred from Bachelor of Science in Mathematics (UNSW 3956).
- Academic standing: WAM 83.5; admitted to the UNSW [Talented Students Program](#) (TSP) in 2025.

RESEARCH INTERESTS

Quantum computing. Quantum algorithms and quantum simulation.

Theoretical computer science. Computational complexity and algebraic algorithms: tensor, group, and graph isomorphism; linear code equivalence (LCE); reductions among equivalence problems.

Algebra. Multilinear algebra and tensor spaces, group actions, canonical forms and invariants.

RESEARCH EXPERIENCE

Research, University of Technology Sydney (UTS)

Sydney, Australia

Supervised by Youming Qiao

October 2026 – Present

- Starting work on the linear code equivalence (LCE) and matrix code equivalence (MCE) problems.

Research, University of Technology Sydney (UTS)

Sydney, Australia

Supervised by Youming Qiao and Murray Elder

Late 2025 – September 2026

- Studied isomorphism and automorphism problems for tensors, groups, algebras, and polynomials.
- Proved that an r -uniform hypergraph is connected if and only if its associated tensor is indecomposable, linking graph structure to tensor structure; contributed to the manuscript below.

Independent Study, University of Technology Sydney (UTS)

Sydney, Australia

Supervised by Youming Qiao

Term 2 – Term 3, 2025

- Supervised reading in quantum computing: Nielsen and Chuang, *Quantum Computation and Quantum Information*, and other standard texts.

Talented Students Program, UNSW

Sydney, Australia

Supervised by Quoc Le Gia

Term 1, 2025

- Supervised reading of selected literature in large language models; practice in extracting technical assumptions and presenting mathematical and algorithmic ideas.

MANUSCRIPTS

- Murray Elder, **Chang He**, Luke Mathieson, Euan Mendoza, and Youming Qiao. *On isomorphism and automorphism problems for tensors, groups, algebras, and polynomials*. Submitted to ITCS 2027, under review (authors in alphabetical order).

TECHNICAL SKILLS

Programming. Python, C/C++, JavaScript/TypeScript; Git, LaTeX, Unix shell.

Quantum and AI. Qiskit (working knowledge); building and using LLM agents.

ACADEMIC ACTIVITIES

- Member, Sydney Algorithms and Computing Theory (SACT) group, University of Sydney.
- [MathQuEST Reading Group on Quantum Computing](#), University of Sydney, 2026.
- [Sydney TCS Winter School](#): Interactive Proofs and the PCP Theorem, University of Sydney, July 2026.